

ANSWER KEY & DETAILED EXPLANATIONS

Adaptive Digital SAT Practice Test
Horizon Series — Form A

Test Structure

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Total: 147 questions. All answers have been verified, and the math solutions have been independently recomputed.

SECTION 1: READING & WRITING

Module 1 — All Students (27 Questions)

Q1	D	Q2	C	Q3	D	Q4	A	Q5	C	Q6	B	Q7	C
Q8	B	Q9	A	Q10	C	Q11	B	Q12	B	Q13	C	Q14	A
Q15	C	Q16	D	Q17	D	Q18	A	Q19	D	Q20	B	Q21	A
Q22	A	Q23	D	Q24	B	Q25	B	Q26	D	Q27	B		

Detailed Explanations

Q1: D

The passage describes data from decades-old expeditions still being used today, indicating the contributions are lasting or "enduring." A) "Isolated" means cut off, contradicting continued use. C) "Unprecedented" means never done before, but the emphasis is on duration, not novelty. B) "Provisional" means temporary, the opposite of the passage's meaning.

Q2: C

Phytoplankton are consumed by organisms higher in the food chain and their decline causes cascading effects, establishing them as essential ("fundamental to") the ecosystem. A) "Tangential to" means only slightly related. B) "Indistinguishable from" doesn't fit logically. D) "Competitive with" implies rivalry, not a foundational role.

Q3: D

The blank needs a word meaning "produce." Works can have as many analyses as there are critics, so the works "generate" many interpretations. A) "Dismiss" means to reject. C) "Constrain" means to limit. B) "Obscure" means to hide.

Q4: A

Languages with very few color terms compared to the global average of eleven indicate vocabularies tend toward being "sparse" (few in number). B) "Rigid" describes inflexibility, not quantity. C) "Uniform" means all the same. D) "Elaborate" means complex, the opposite.

Q5: C

The text presents two institutions, each with its specific focus area. A) No historical development is traced from one to the other. B) Funding models aren't discussed. D) Collection sizes aren't contrasted.

Q6: B

The poem contrasts the speaker's harsh working life ("drudge," "jostling crowd") with the river's peace ("quiet place," "silent dream"). A) Exercise isn't mentioned. C) No danger is described. D) Only one river appears.

Q7: C

The passage states non-community-controlled museums "tend to anticipate primarily non-Indigenous visitors." A) Artifact numbers aren't compared. B) Physical size isn't discussed. D) Historical vs. contemporary focus isn't mentioned.

Q8: B

The passage states the cormorant "adapted to forage in the nutrient-rich coastal waters," indicating adaptation to a particular habitat. A) Not the deepest waters -- coastal waters. C) Habitat sharing isn't the reason given. D) Threats explain vulnerability, not specialization.

Q9: A

The narrator shifts from a pressing, confined feeling ("seemed to press in upon me") to openness as the prairie opens up, paralleling unease to spaciousness. B) No frustration about journey length is expressed. C) The trip appears planned. D) The narrator distinguishes between town and prairie.

Q10: C

In 1995: Class A is roughly 40, Class B roughly 30, Class C roughly 35 hectares. All three values fall between approximately 30 and 45 hectares. A) 100-150 describes 1975 values. B) 60-80 describes values around 1985. D) 140-180 only matches one series in 1975.

Q11: B

The hoverfly shows approximately 0 observations in both March and December, making it the species that was entirely absent at those times. A) The meadow brown has roughly 5 in June, not zero. C) The damselfly has roughly 12 in September. D) The meadow brown has roughly 5 in December, not zero.

Q12: B

If water near Palau is warmer at any given depth, then even though Okinawan fish develop at shallower (warmer) depths, Palauan fish could still be in equally warm or warmer water, directly weakening the hypothesis. A) Post-juvenile behavior doesn't address the temperature comparison. C) and D) would support or strengthen the hypothesis.

Q13: C

The positive nitrogen-diversity correlation appears globally (Japan and elsewhere) but not in Scandinavia, suggesting something specific to Scandinavian conditions disrupted a general pattern. A) Doesn't explain the missing correlation. B) Contradicts the data. D) Evidence from Japan shows positive associations outside Europe.

Q14: A

Despite different political structures, crop types, and geographic constraints across regions, the same scaling pattern emerged everywhere. This implies that inter-regional differences had limited effect on irrigation patterns. B) Contradicted by cross-regional evidence. C) Not supported by the passage. D) The passage states geographic constraints were "regionally distinct," not shared.

Q15: C

"They" correctly refers to the plural committee members who reached a consensus. A) "Anyone" is singular/indefinite. B) "You" shifts to second person. D) "It" is singular and refers to things.

Q16: D

"It" correctly refers to the singular "the television program." A) "Those" is plural and demonstrative. B) "They" is plural. C) "Each" doesn't make sense for a single show.

Q17: D

The sentence beginning with "we recognize that" is declarative, requiring a period. D provides the correct subject-verb-object structure with a period. A ends with a comma, B uses a question mark, and C inverts subject/verb incorrectly.

Q18: A

A comma after "Europe" correctly continues the introductory dependent clause. B) A semicolon needs two independent clauses, but "While..." makes the first clause dependent. C) A period creates a fragment. D) No punctuation creates a run-on.

Q19: D

"Anthems' lengths" uses the correct plural possessive form. Two anthems are discussed, and "lengths" (belonging to the anthems) is the subject of "correlate." A) "Anthems' length" uses singular "length" despite two anthems. B) "Anthems lengths" omits the required possessive apostrophe. C) "Anthem's" is singular possessive.

Q20: B

No punctuation is needed -- "American artist Edward Hopper" is a single noun phrase where "artist" is a title preceding the name. A) Colon incorrectly separates descriptor from name. C) Dash creates unnecessary interruption. D) Comma incorrectly separates title from name.

Q21: A

"Often" correctly describes a general tendency (frequency) of years elapsing between selection and recognition. B) "Therefore" implies a logical cause-effect conclusion. C) "Similarly" needs a parallel comparison. D) "Indeed" merely affirms a point.

Q22: A

"In other words" rephrases the broader principle: the same social coordination mechanism that can produce benefits (starlings avoiding hawks) also makes birds susceptible to behavioral errors (geese following wrong corrections). B) "Conversely" implies a full reversal of topic. C) "Indeed" simply affirms. D) "Consequently" implies direct causation.

Q23: D

Option D describes the process of oil painting (pigments combined with drying oil, applied to canvas) without extraneous details about Hopper or fresco. A) Compares techniques rather than describing the oil painting process. B) Focuses on Hopper. C) Only identifies a work.

Q24: B

B clearly shows the contrast: Malaysia's tallest building IS in its capital, while Australia's is NOT. A) Incorrectly states both are not in their capitals. C) Lists facts without making a comparison. D) Only names capitals without discussing buildings.

Q25: B

B explains Nazario's documentation method: retracing individual journeys in personal detail. A) Only names a country. C) Focuses on awards, not methodology. D) Mentions Enrique but doesn't explain the documentation approach.

Q26: D

D highlights what both musicians share: contributing their musical talents to Yo-Yo Ma albums. A) Focuses on timing differences. B) Describes albums without connecting the musicians. C) Emphasizes a difference (instruments), not a similarity.

Q27: B

B generalizes that early appearances were modest and low-profile, and supports this with the specific example of limited play with the Montreal Royals. A) Chronological but doesn't characterize the career. C) Doesn't focus on early career. D) Provides a fact without a generalization.

SECTION 1: READING & WRITING

Module 2 — Easier Path (27 Questions)

Q1	D	Q2	B	Q3	D	Q4	C	Q5	A	Q6	C	Q7	B
Q8	A	Q9	B	Q10	B	Q11	C	Q12	B	Q13	A	Q14	A
Q15	A	Q16	D	Q17	B	Q18	D	Q19	B	Q20	C	Q21	D
Q22	A	Q23	D	Q24	B	Q25	C	Q26	B	Q27	B		

Detailed Explanations

Q1: D

"Address" means to deal with or tackle a problem. The advocates closed the data gap by partnering with researchers.

Q2: B

Laboratory findings "confirm" the previously suggested idea, providing direct supporting evidence.

Q3: D

The passage sets up a contrast: not torpor (sluggishness) but focused alertness, i.e., "attentiveness."

Q4: C

Challenges and emphasis on other species have resulted in "a scarcity" (lack) of nighttime vocalization data.

Q5: A

The underlined sentence says horizontal transmission is more prevalent than assumed, implying the common understanding may be inaccurate.

Q6: C

The text describes a characteristic (expansive scope) and presents two differing viewpoints: critics who see it as meandering vs. those who see it as intricately woven.

Q7: B

The text first discusses Cash's mid-1950s/early 1960s work, then looks back to an earlier period (gospel singing in Arkansas a decade earlier).

Q8: A

Osei's research shows that incidental activity (not just structured workouts) improves mood, suggesting the framework overlooks this beneficial category.

Q9: B

Both texts discuss scientific efforts to understand marsupials -- Text 1 notes the difficulty, Text 2 describes new sensor technology overcoming it.

Q10: B

The passage states a 150,000 kg blue whale could consume 2,250 kg of krill vs. 450 kg for a 30,000 kg humpback, showing the blue whale can consume more.

Q11: C

U1's correlation for classical (roughly 0.38) is higher than for electronic (roughly 0.22), so the model predicted U1 would derive more satisfaction from classical music.

Q12: B

Nora's anguished self-address ("Corrupt my own children? Poison my home?") most directly illustrates her increasingly unbearable tension about the secret she conceals.

Q13: A

If earthworms accumulate TiO₂-NPs with fewer negative health effects than moles, earthworm monitoring wouldn't accurately proxy for the health impacts measured in vertebrates, weakening the hypothesis.

Q14: A

Brief, intense light bursts being especially effective at triggering startle reflexes directly supports the claim that bioluminescence in ink-releasing species functions to startle predators.

Q15: A

Only bead-drilling declined while other traditions continued, suggesting demographic continuity (no population displacement) and that the bead-drilling decline had a specific, different cause.

Q16: D

Plural subject "observations" takes the plural verb "illuminate" (not singular "illuminates").

Q17: B

The sentence opens a parenthetical with an em dash before "including" and must close it with another em dash after "Hephaestus" to properly bracket the list before continuing to the main clause.

Q18: D

"Is referred" is the correct passive voice form: a metrical foot [that] is referred to as an anapest.

Q19: B

The comma before "and" correctly joins two independent clauses with a coordinating conjunction.

Q20: C

"Electricity, though" correctly introduces a concessive subordinate clause without additional punctuation that would fragment the sentence.

Q21: D

"Therefore" signals a logical result: the album received widespread praise; therefore, few were surprised it won a Grammy.

Q22: A

"Still" introduces a concession: despite the seemingly imprecise method (plaster model and balancing pin), the result was remarkably accurate.

Q23: D

"Conversely" signals an opposing case: ekphrastic novels typically engage with visual artwork, but Cunningham's conversely uses a Whitman poem.

Q24: B

B provides a specific example: the 1877 production of Swan Lake as a particular production choreographed by Julius Reisinger.

Q25: C

C provides a specific example: China uses the yuan as the basic unit of its national currency, the renminbi.

Q26: B

B emphasizes the similarity: both mass fraction and mole fraction indicate relative abundance of elements, despite measuring different quantities.

Q27: B

B generalizes that Hadid's earliest works were largely theoretical and unrealized, supported by the specific example of the 1983 Peak competition entry.

SECTION 1: READING & WRITING

Module 2 — Harder Path (27 Questions)

Q1	D	Q2	D	Q3	A	Q4	A	Q5	C	Q6	A	Q7	A
Q8	A	Q9	C	Q10	D	Q11	B	Q12	C	Q13	A	Q14	D
Q15	C	Q16	A	Q17	A	Q18	D	Q19	B	Q20	D	Q21	A
Q22	C	Q23	D	Q24	A	Q25	A	Q26	B	Q27	D		

Detailed Explanations

Q1: D

"Assimilated" means absorbed or incorporated. The architects incorporated local Slavic building traditions into their designs, not rejecting, cataloguing, or replacing them.

Q2: D

"Tenuous" means weak or slight. Empirical support remains weak because studies poorly replicate natural forest conditions.

Q3: A

Chee builds compositions from Navajo chant melodies, making Indigenous identity more "legible" (recognizable) than in Tate's post-minimalist approach where few listeners would recognize Indigenous origins.

Q4: A

Only optical patents declined while chemical synthesis continued, suggesting the broader engineering workforce remained intact and the optical disruption was specific to that field.

Q5: C

An unchecked Escovopsis infection would destroy the entire garden, permanently eliminating the colony's sole food source, making the short-term costs of weeding worthwhile.

Q6: A

If cities with comparable transit improvements but no congestion pricing saw similar ridership increases, the pricing itself isn't the direct cause of the ridership increase.

Q7: A

Joye (Text 2) would argue the improvements may stem from reduced sensory complexity and lower noise, not specifically from the natural character of the environments.

Q8: A

Bilingual speakers using the dominant language at work while maintaining the minority language only at home directly demonstrates "prestige asymmetry" even where legal protections exist.

Q9: C

Adding 7 to each y-value shifts the entire line of best fit up by 7. The original y-intercept is roughly 58, so the new y-intercept is $58 + 7 = 65$.

Q10: D

A colon introduces an explanation or elaboration of the epistemological "tension" described. A semicolon would merely separate two independent clauses without signaling elaboration.

Q11: B

The singular subject "discovery" takes "has," not "have." The intervening phrases ("of high arsenic concentrations... alongside subsequent studies...") are parenthetical and do not change the subject.

Q12: C

"However" introduces a contrast: the simplified narratives are psychologically comforting but can impede reconciliation.

Q13: A

The finding that plasticity can be pharmacologically reopened suggests critical-period mechanisms are more modifiable than the traditional model implies, potentially opening therapeutic avenues.

Q14: D

"For example" introduces a specific illustration of the general finding: a consumer who would pay 20% more might pay only 8% more when carbon footprint information is also shown.

Q15: C

"Architects'" is the correct plural possessive: the palaces belonging to three architects. A) singular possessive. B) no possessive. D) nonstandard form.

Q16: A

During rapid transitions, moisture source changes can affect the isotope ratio, making temperature reconstructions for rapid-change periods less reliable than those for stable periods.

Q17: A

If peregrines increased at comparable rates in rural areas without pigeons but with other similar-sized prey, pigeon abundance specifically isn't the primary driver.

Q18: D

D connects Kere's philosophy (community participation, local materials) to a specific project: villagers making compressed laterite clay bricks on-site and contributing their labor at Gando Primary School.

Q19: B

B presents both sides of the debate: critics argue key factors remain essentially unknown, while supporters maintain the equation provides a useful framework for scientific inquiry.

Q20: D

A semicolon correctly separates two independent clauses: "code-switching does not reflect linguistic confusion" and "it instead demonstrates sophisticated metalinguistic awareness."

Q21: A

"Has involved" maintains parallelism with "began" and uses the present perfect correctly for an ongoing action: the initiative began in 2018 and has involved annual expeditions since then.

Q22: C

"Taken together" summarizes the cumulative findings (nuanced emotional responses to unfamiliar scales) that collectively challenge the view that emotional responses to music are primarily shaped by cultural familiarity.

Q23: D

"The significance of the find was immediately recognized:" connects the participial phrase ("Discovered by... Mary Anning...") to a grammatically correct main clause about the skeleton's significance.

Q24: A

The dual system (written statutes for commerce, customary law for family/land) shows that the imposition of written codes was more selective than a simple narrative of cultural displacement would imply.

Q25: A

Two distinct chemical abundance patterns within the same cluster provide direct evidence of successive generations formed from progressively enriched gas, supporting sequential self-enrichment.

Q26: B

B directly connects the technical details (30+ interlocking gears, eclipse prediction, planetary tracking) to the historical significance: this level of complexity was unmatched for a millennium.

Q27: D

"Nevertheless" introduces a contrast: despite the instruments far outlasting their projected lifespan, the plutonium-238 generators are gradually losing power output.

SECTION 2: MATH

Module 1 — All Students (22 Questions)

Q1	B	Q2	18	Q3	C	Q4	D	Q5	B	Q6	C	Q7	C
Q8	D	Q9	64	Q10	A	Q11	56	Q12	C	Q13	B	Q14	A
Q15	D	Q16	C	Q17	A	Q18	A	Q19	20	Q20	B	Q21	245
Q22	6												

Detailed Explanations

Q1: B

Perimeter = $a + b + a + b = 2a + 2b = 58$.

Q2: 18

$5 + x = 3$, so $x = -2$. Then $30 + 6(-2) = 30 - 12 = 18$.

Q3: C

Almonds: $3 \times 160 = 480$ cal. Cranberries total: $1,740 - 480 = 1,260$. Per bag: $1,260 / 5 = 252$.

Q4: D

15% of \$180 = $0.15 \times 180 = \$27$.

Q5: B

8 servings need 3 cups. 24 servings = 3×8 servings, so $3 \times 3 = 9$ cups.

Q6: C

$5x = 20$, so $x = 4$. Then $-2(4) + y = -3$, so $y = 5$. $x + y = 4 + 5 = 9$.

Q7: C

Cost minus credit: $1.25n - 3$. Must be at most \$15: $1.25n - 3 \leq 15$.

Q8: D

Area = $\pi \times r^2 = \pi \times 24^2 = 576\pi$.

Q9: 64

Radius of M = $\sqrt{25} = 5$. Radius of N = $5 + 3 = 8$. So $p = 8^2 = 64$.

Q10: A

Combine like terms: $5ab^2 + 2ab^2 = 7ab^2$; $3ab$ stays. Result: $7ab^2 + 3ab$.

Q11: 56

$6a/(4b) = 84$. So $6a = 336b$, meaning $a = 56b$. Therefore $a/b = 56$.

Q12: C

Similar triangles have equal corresponding angles. Angle PQR corresponds to angle PST = 63 degrees. The side ratio (15x) does not affect angles.

Q13: B

The point (12.5, 24.8) on the graph means $x = 12.5$ (carbon atoms) corresponds to $y = 24.8$ (melting point in degrees C).

Q14: A

Slope = $(22 - 10)/(5 - 2) = 12/3 = 4$. $g(x) = 4x + b$. $g(2) = 8 + b = 10$, so $b = 2$. $g(0) = 0 + 2 = 2$.

Q15: D

$90/x = 5x$. Multiply both sides by x : $90 = 5x^2$. So $5x^2 = 90$.

Q16: C

Check (7, 7): $7 \leq 3(7) - 14 = 21 - 14 = 7$. Since $7 \leq 7$ is true, (7, 7) is a solution.

Q17: A

$x + y \leq 12$ (total bulbs at most 12) and $2x + 3y \leq 30$ (tulips need 2 sq in, daffodils 3 sq in). This is system I.

Q18: A

$h = (5r)^2/(3s) = 25r^2/(3s)$. So $3hs = 25r^2$, $r^2 = 3hs/25$, $r = \sqrt{3hs}/5$.

Q19: 20

Line m has slope 5 (parallel to $y = 5x + 3$) and passes through (0, 0): $y = 5x$. At $x = 4$: $d = 5(4) = 20$.

Q20: B

Standard form: $(x - h)^2 + (y - k)^2 = r^2$. Center (11, 23), radius 9k. $r^2 = (9k)^2 = 81k^2$.

Q21: 245

$\tan(\theta) \leq 1/10$. Since $\tan(\theta) = \text{height}/x = 24.5/x$, the minimum x satisfies $24.5/x = 1/10$, so $x = 245$.

Q22: 6

The curve passes through (0, -3): $a^0 + b = 1 + b = -3$, so $b = -4$. Asymptote $y = b = -4$. From the graph, $a = 2$. So $a - b = 2 - (-4) = 6$.

SECTION 2: MATH

Module 2 — Easier Path (22 Questions)

Q1	C	Q2	B	Q3	D	Q4	C	Q5	C	Q6	A	Q7	A
Q8	D	Q9	270	Q10	C	Q11	D	Q12	C	Q13	A	Q14	7
Q15	B	Q16	A	Q17	B	Q18	-6	Q19	D	Q20	B	Q21	3
Q22	720												

Detailed Explanations

Q1: C

$$f(-4) = (-4 + 12)/4 = 8/4 = 2.$$

Q2: B

$$(15 + 18 + 20 + 22 + 24 + 25 + 30)/7 = 154/7 = 22.$$

Q3: D

10% of 20 degrees C = 2 degrees C. From the graph, this occurs at approximately $t = 2.3$ minutes.

Q4: C

Adding 5 to each y-value shifts the y-intercept up by 5: $8 + 5 = 13$.

Q5: C

$g(0) = 8$ and $g(4) = 8$, with both outputs the same, so g is a constant function: $g(x) = 8$.

Q6: A

$f(x) = -5(x - 2) = -5x + 10$. Check: $f(2) = 0$, $f(6) = -20$, $f(12) = -50$. All match.

Q7: A

$m(45) = 200(0.5)^3 = 200/8 = 25$. Interpretation: approximately 25 grams remains 45 years after decay starts.

Q8: D

$$P(\text{HR} \mid \text{NOT Engineering}) = P(\text{HR}) / P(\text{NOT Eng}) = 0.15 / (1 - 0.40) = 0.15/0.60 = 0.25.$$

Q9: 270

$$V = (1/3) \times \pi \times r^2 \times h = (1/3) \times \pi \times 9^2 \times 10 = (1/3)(810\pi) = 270\pi. \text{ So } k = 270.$$

Q10: C

By the midsegment theorem, ST is parallel to PR and $ST = PR/2$, so $PR = 2 \times 6 = 12$. ($PT = 11$ is a distractor.)

Q11: D

$x(r - 5) + 4 = 17x + 30$ simplifies to $x(r - 22) = 26$. For exactly one solution, $r - 22$ cannot equal 0. If $r = 22$, then $0 = 26$, which is impossible. So r CANNOT be 22.

Q12: C

20% green = 30 marbles, so total = 150. Blue = $40\% \times 150 = 60$.

Q13: A

$11x^2 + 23x + (c - 7) = 0$. Discriminant: $23^2 - 4(11)(c - 7) = 837 - 44c$. Setting discriminant = 309: $837 - 44c = 309$, so $44c = 528$, $c = 12$. Solution: $(-23 + \sqrt{309})/22$ confirms.

Q14: 7

$3x^3 + 45x^2 + 168x = 3x(x^2 + 15x + 56) = 3x(x + 7)(x + 8)$. Factors include $(x + 7)$ and $(x + 8)$. Smallest positive $d = 7$.

Q15: B

"Increases 150%" means the population is multiplied by 2.5 each year. At $t = 1$: initial $\times 2.5 = 375$, so initial = 150. Equation: $n = 150(2.5)^t$.

Q16: A

Garden 20×15 ft with border of width w . Total area: $(20 + 2w)(15 + 2w)$. Border = total - 300 = 200, so total = 500. With $w = 2.5$: $(25)(20) = 500$. Confirmed.

Q17: B

Check $2x + ty = 10t$. With $(-3t, 16)$: $2(-3t) + t(16) = -6t + 16t = 10t$. With $(-t, 12)$: $-2t + 12t = 10t$. With $(t, 8)$: $2t + 8t = 10t$. All check out.

Q18: -6

Five consecutive even integers: $x, x+2, x+4, x+6, x+8$. Condition: $36(x + 4) \leq (x + x + 8) - 60$. Simplifying: $36x + 144 \leq 2x - 52$, so $34x \leq -196$, $x \leq -5.76$. Greatest even integer: -6.

Q19: D

Original: 24 inches represents $24 \times 12 = 288$ feet. 50% reduction: $24 \times 0.5 = 12$ inches. 12 inches still = 288 feet, so 1 inch = $288/12 = 24$ feet.

Q20: B

$q(x) = a(3.2)^x + b$. At $x = 0$: $q(0) = a + b$, with both constants directly visible. $p(x) = a(3.2)^{(x+b)} = a(3.2)^b \times (3.2)^x$, so $p(0) = a(3.2)^b$, which is not a directly readable constant. Answer: q only.

Q21: 3

$k(x) = -|x| + bx + c$. From $(0, -\sqrt{195})$: $c = -\sqrt{195}$. From $(3, 0)$: $-3 + 3b - \sqrt{195} = 0$, so $b = (3 + \sqrt{195})/3$. For $m \geq 0$: $m(b - 1) + c = 0$, so $m = \sqrt{195} / (\sqrt{195}/3) = 3$.

Q22: 720

$g(x) = |x|/a - 18$, $a < 0$. $g(12a) = |12a|/a - 18 = -12a/a - 18 = -12 - 18 = -30$. $g(6a) = -6 - 18 = -24$. Product: $(-30)(-24) = 720$.

SECTION 2: MATH

Module 2 — Harder Path (22 Questions)

Q1	C	Q2	B	Q3	D	Q4	C	Q5	C	Q6	C	Q7	A
Q8	C	Q9	900	Q10	C	Q11	D	Q12	A	Q13	6	Q14	B
Q15	D	Q16	B	Q17	-9	Q18	D	Q19	B	Q20	5	Q21	864
Q22	36												

Detailed Explanations

Q1: C

$g(-3) = (2(-3) + 16)/5 = (-6 + 16)/5 = 10/5 = 2$.

Q2: B

$(12 + 14 + 17 + 19 + 21 + 22 + 28)/7 = 133/7 = 19$.

Q3: D

25% of 800 = 200 becquerels. From the graph, this occurs at approximately $t = 2.8$ hours.

Q4: C

Subtracting 11 from each y-value shifts the y-intercept down by 11: $72 - 11 = 61$.

Q5: C

$h(0) = 12$, $h(6) = 12$ -- both outputs are 12, so h is a constant function: $h(x) = 12$.

Q6: C

$f(x) = -6(x + 2) = -6x - 12$. Check: $f(3) = -6(5) = -30$, $f(7) = -6(9) = -54$, $f(14) = -6(16) = -96$. All match.

Q7: A

$d(120) = 500(0.5)^3 = 500/8 = 62.5$. Interpretation: approximately 62.5 grams remains 120 years after decay starts.

Q8: C

$P(\text{Senior} \mid \text{at least Junior}) = P(\text{Senior}) / (P(\text{Junior}) + P(\text{Senior})) = 21 / (24 + 21) = 21/45 = 0.467$, closest to 0.47.

Q9: 900

$V = \pi \times r^2 \times h = \pi \times 10^2 \times 9 = 900\pi$. So $k = 900$.

Q10: C

By the midsegment theorem, MN is parallel to XZ and $MN = XZ/2$, so $XZ = 2 \times 11 = 22$. ($MY = 18$ is a distractor.)

Q11: D

$x(r - 8) + 5 = 23x + 40$ simplifies to $x(r - 31) = 35$. If $r = 31$, then $0 = 35$, which is impossible. So r CANNOT be 31.

Q12: A

$15x^2 + 31x + (c - 8) = 0$. Discriminant: $31^2 - 4(15)(c - 8) = 1441 - 60c$. Setting discriminant = 481: $1441 - 60c = 481$, so $60c = 960$, $c = 16$. Solution: $(-31 + \sqrt{481})/30$ confirms.

Q13: 6

$4x^3 + 52x^2 + 168x = 4x(x^2 + 13x + 42) = 4x(x + 6)(x + 7)$. Factors include $(x + 6)$ and $(x + 7)$. Smallest positive $d = 6$.

Q14: B

"Increases 120%" means the value is multiplied by 2.2 each year. At $t = 1$: initial $\times 2.2 = 440$, so initial = 200. Equation: $v = 200(2.2)^t$.

Q15: D

Photo 16×12 with frame width w . Total area: $(16 + 2w)(12 + 2w)$. Frame area = total - $192 = 288$, so total = 480. With $w = 4$: $(24)(20) = 480$. Confirmed.

Q16: B

Check $3x + ty = 6t$. With $(-4t, 18)$: $3(-4t) + t(18) = -12t + 18t = 6t$. With $(-2t, 12)$: $-6t + 12t = 6t$. With $(2t, 0)$: $6t + 0 = 6t$. All check out.

Q17: -9

Six consecutive odd integers: $x, x+2, x+4, x+6, x+8, x+10$. Condition: $30(x + 6) \leq (x + x + 10) - 42$. Simplifying: $30x + 180 \leq 2x - 32$, so $28x \leq -212$, $x \leq -7.57$. Greatest odd integer: -9.

Q18: D

Original: 40 inches represents $40 \times 20 = 800$ feet. "70% shorter" means 30% of original: $40 \times 0.3 = 12$ inches. 12 inches = 800 feet, so 1 inch = $800/12 = 66.67$ feet.

Q19: B

$g(x) = a(4.5)^x + b$. At $x = 0$: $g(0) = a + b$, with both constants directly visible. $f(x) = a(4.5)^{(x+b)}$ yields $f(0) = a(4.5)^b$, which is not a simple constant. Answer: g only.

Q20: 5

$j(x) = -|x| + bx + c$. From $(0, -\sqrt{510})$: $c = -\sqrt{510}$. From $(5, 0)$: $-5 + 5b - \sqrt{510} = 0$, so $b = (5 + \sqrt{510})/5$. For $m \geq 0$: $m = \sqrt{510} / (\sqrt{510}/5) = 5$.

Q21: 864

$f(x) = |x|/a - 22$, $a < 0$. $f(10a) = |10a|/a - 22 = -10 - 22 = -32$. $f(5a) = -5 - 22 = -27$. Product: $(-32)(-27) = 864$.

Q22: 36

$3x^2 - 7x - 20 = 0$. Discriminant: $49 + 240 = 289 = 17^2$. $x = (7 + 17)/6 = 24/6 = 4$ (taking $x > 0$). So $9x = 36$.